Order Effects Without Attractors:
A Pre-Registered Architecture Ladder for Memory-Augmented LLM Agents

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Abstract

Deployed language-model agents increasingly carry persistent memory, and a growing literature documents behavioural *drift* over long interactions. We ask what *governs* it. Under a pre-registered protocol with decision rules frozen before data collection, we compare a ladder of memory architectures—memoryless, raw append-only log, self-summarising recurrent memory, and semantic vector memory—against an engineered agent whose explicit low-dimensional state is coupled to *perception*. Two properties usually conflated as "drift" separate cleanly. **Order-dependence** is governed by retrieval addressing: content-addressed memory transmits early history to the behavioural endpoint ($|r| = 0.75$, 99.5th percentile of a matched random-walk null), whereas recency-windowed and self-rewriting memories erase it (3.7th and 78.2nd). Retrieval logging exhibits the mechanism directly: early self-exemplars still occupy 58% of retrieved slots after a full corrective burst, 1.36× their positional base rate. **Attractor structure** is not: every natural architecture is monostable and correctable across three models including an ablated one, and behavioural correction *outpaces* store composition; only the engineered perception-coupled state yields wide, bimodal, hysteretic basins. Finally, an unpredicted observation with direct deployment relevance: the least cautious agent is the one that has experienced nothing at all—after thirty-five mundane cooperative turns, agents complied outright on half of a safeguard battery, while memory of social pressure in *either* direction raised adherence. We conclude that structured behavioural drift is architectural rather than an emergent consequence of memory persistence.

1 Introduction

A language-model agent with persistent memory conditions each response on a record of what came before. The natural expectation is that this makes behaviour *path-dependent*: early interactions should disproportionately shape later conduct, and the agent should be able to settle into stable behavioural modes that resist correction. This expectation underlies much of the concern about long-horizon agent deployment, where an assistant may accumulate hundreds of turns of history before taking a consequential action.

We test that expectation directly and largely overturn it. Our central observation is that two phenomena routinely bundled together under the word "drift" are in fact separable and have different architectural causes:

1. **Order-transmission**—whether *when* something happened leaves a trace in the final behaviour—is governed by how memory is *addressed*.
2. **Attractor formation**—whether behaviour settles into wide, stable, correction-resistant basins—does not appear in any natural memory architecture we tested, and requires an explicit low-dimensional state coupled to perception.

Contributions. (i) A pre-registered architecture ladder isolating retrieval addressing as the operative dimension for order effects (§4). (ii) Direct measurement, at the retriever, of the mechanism that transmits early history—and evidence that transmission does not imply lock-in (§5). (iii) An unpredicted, deployment-relevant observation: unpressured cooperative memory is the *least* cautious state (§6). (iv) A cheap "appraisal-field" method, and the diagnostic value of its failure on natural agents (§7).

2 Related work

Persona and identity drift. Li et al. [1] introduced a quantitative benchmark for persona stability using self-chats between personalised chatbots, reporting significant drift within eight rounds on LLaMA2-chat-70B and attributing it in part to attention decay over long exchanges. Choi et al. [2] examined identity consistency across nine models and found that larger models experienced greater identity drift, and that assigning a persona did not reliably prevent it. Most relevant to our setting, ContextEcho [3] measured persona drift at deployment scale in long agentic-coding sessions, introducing a snapshot-then-probe protocol that forks conversation state without perturbing the main session—the same measurement discipline we adopt (§4).

Memory as a source of behavioural change. Dabas et al. [4] studied *memory-induced tool-drift*: biases stored in memory silently affect tool calls in contexts where they do not apply. They report that biased memories act as implicit steering vectors, and that standard defences reduce but do not eliminate the effect. Wei et al. [5] describe a self-reinforcing error cycle in which a corrupted outcome is stored as precedent, amplifying the initial error and progressively lowering the threshold for similar failures. Lin et al. [6] survey long-term memory security and reframe the governing question from whether the current input is harmful to what memory *state* the system is in.

Position of this work. That literature is predominantly adversarial or measurement-oriented: it studies injected or poisoned memory, and measures drift in black-box systems. Our contribution is complementary in two ways. First, the self-precedent effect we observe arises with *no adversary*—it is produced by ordinary interaction. Second, we run a controlled comparison *across memory architectures* under an identical protocol, which localises which architectural property produces which phenomenon. In particular, the self-reinforcing precedent described adversarially by [5] appears in our setting as transmission *without* lock-in: correction outpaces store composition (§5).

3 An engineered state exhibits the full drift signature

We first establish that a memory-conditioned LLM system *can* exhibit the complete signature, so that a null result on natural agents is informative rather than an artefact of a weak protocol.

3.1 Model

The agent carries a state on two timescales. A fast *mood* (v, r) relaxes towards the appraisal of the current event, and a slow *trait* vector—of which we track the attachment axis $x_A \in [-1, 1]$—integrates sustained mood. Writing $\hat{v}_t$ for the perceived valence of event t,

$$v_{t+1} = v_t + k_v\big(\hat{v}_t - v_t\big), \qquad r_{t+1} = r_t + k_r\big(\hat{a}_t - r_t\big), \tag{1}$$

with k_v, k_r large enough that mood equilibrates within a few interactions. The trait axis follows a cubic (double-well) update,

$$\Delta x_A \;=\; \big(a\,x_A - b\,x_A^3 + w\,v_t\big)\,\eta(t), \tag{2}$$

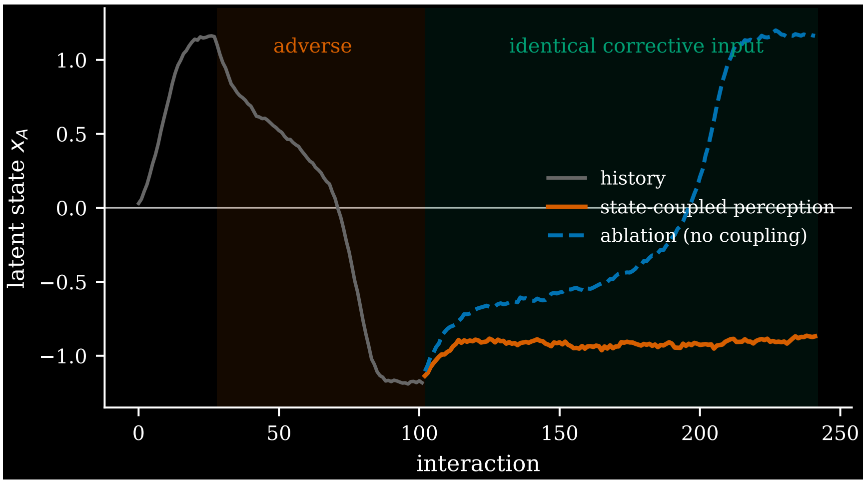

Figure 1: The engineered signature. A state captured in the adverse basin does not recover under corrective input of equal magnitude when perception is state-coupled (orange), while the ablated system recovers fully (blue). Plasticity is held constant, so the asymmetry is attributable to Eq. (4) rather than to annealing.

whose deterministic part has fixed points at $x_A^* = 0$, unstable, and $x_A^* = \pm\sqrt{a/b}$, stable. The cubic term saturates the dynamics at the wells, so the state cannot run away; the unstable centre means the system cannot remain neutral. Plasticity anneals with age,

$$\eta(t) \;=\; \eta_0 \exp(-t/\tau), \tag{3}$$

so that early events act while the learning rate is high—path-dependence by construction, and the computational analogue of a critical period.

3.2 Perception coupling

The load-bearing component is that state biases *interpretation*, not merely expression. Given raw appraisal primitives—care c, threat θ, novelty ν, autonomy α—the perceived care of an event is

$$\tilde{c} \;=\; c \;-\; \kappa \max(0, -x_A)\,\big(1 - |c|\big), \tag{4}$$

so that a damaged state ($x_A < 0$) reads *ambiguous* events ($|c|$ small) negatively, while unambiguous events are largely unaffected. Perceived valence and arousal follow

$$\hat{v} = \mathrm{clip}\big(\tilde{c} - 0.5\,\theta + 0.3\,\alpha,\ -1,\ 1\big), \qquad \hat{a} = \mathrm{clip}\big(0.7\,\theta + 0.5\,\nu,\ 0,\ 1\big). \tag{5}$$

Equation (4) closes a loop: state $\to$ interpretation $\to$ state. We verified that a real language model reproduces this coupling when supplied with the state and asked to appraise events from the agent's point of view: on ambiguous events, injecting a damaged state versus a secure one shifted appraised care by -0.737 (95% CI $[-0.865, -0.609]$), against a pre-committed prediction of a negative shift in $[-0.6, -0.15]$.

3.3 Signature

Figure 1 shows the consequence. A state driven into the negative basin does not return under corrective input of equal magnitude, because Eq. (4) discounts precisely the input that would restore it; with the coupling ablated, the same input restores the state. Irreversibility is therefore not an added rule—it falls out of the same mechanism that produces the basins.

Table 1: The ladder. Order-transmission is the null percentile of the observed early-history correlation; attractor structure is the spread of endpoints across the frozen orderings. R4 uses a different substrate; its spread is not directly comparable in absolute terms and its status is imported from separately established path-dependence.

rung	addressing	$\|r\|$	null pct.	σ(endpoint)	beyond null
R0	—	0.575*	—	0.071	no
R1	recency	0.015	3.7	0.046	no
R2	rewrite	0.389	78.2	0.051	no
R3	content	**0.751**	**99.5**	0.062	**yes**
R4	(engineered)	0.510	—	**0.764**	yes

* generation noise on a permanently empty store; its spread is the instrument noise floor.

4 The memory ladder

We then ask whether a *natural* memory-augmented agent—retrieve, respond, write a self-authored note, with no engineered scalar anywhere—exhibits the same signature.

4.1 Protocol

All rungs run an identical protocol: the same agent loop, the same fixed judge, the same behavioural probe battery on a caution / safeguard-adherence axis, and the same frozen set of 12 interaction orderings drawn from a single fixed multiset of pressure events. Probes execute in a cloned, discarded context so that measurement cannot mutate the store. The rungs are

	architecture	addressing	state
R0	memoryless	—	none
R1	raw append-only log	recency window	unbounded text
R2	self-summarising memory	rewrite	compressed text
R3	semantic vector store	content (top-k)	unbounded text
R4	engineered latent state	—	low-dimensional, perception-coupled

Path-dependence is assessed against each rung's *own* random-walk null: we generate 10^4 walks with step variance matched to that rung's measured probe increments, and ask whether the observed correlation r between the early-history composition and the final behavioural position exceeds the 95th percentile of $|r|$ under the null.

4.2 Result

Table 1 and Figure 2 give the result, and it is not the one we predicted. Our pre-registered hypothesis held that order effects would track *perception-coupling* specifically. The sharp test was R3: a sophisticated memory with no coupling whatsoever, predicted blind to show no order-dependence. It was the *only* natural architecture to exceed its null, at the 99.5th percentile. The prediction was wrong and is reported as such.

The separating dimension is neither compression, recurrence, nor coupling—the three properties we froze—but **retrieval addressing**. A recency window forgets early turns by construction (3.7th percentile); a self-rewriting summary overwrites its own past (78.2nd); a content-addressed store, having neither decay nor rewrite, remains able to reach arbitrarily far back (99.5th). That the frozen axis did not contain the operative dimension is itself a reportable finding about the axis.

Crucially, R3 passes the order test while showing *no* attractor structure. Its endpoint spread (0.062) lies inside the instrument noise band, all twelve runs finish in the upper region of the

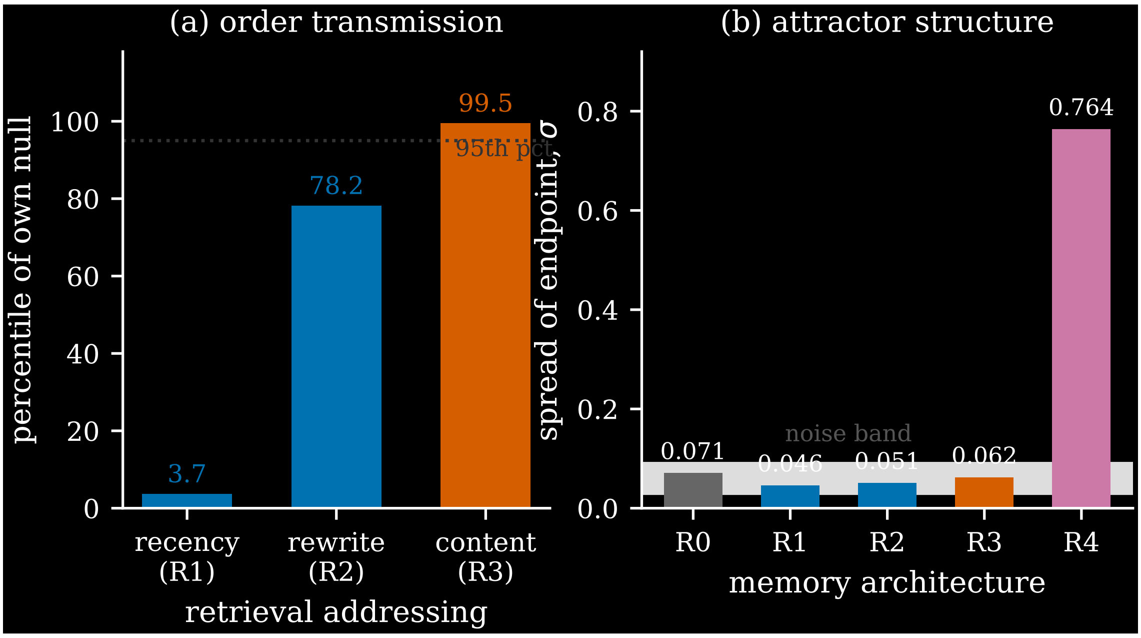

Figure 2: Order-transmission and attractor structure are separable. (a) Order-transmission rises monotonically with retrieval addressing; only content-addressing exceeds the null. (b) Endpoint spread: every natural architecture, including R3, lies inside the instrument noise band; only the engineered perception-coupled state (R4, purple) produces wide basins.

axis, and no bimodality is present—against an engineered spread of 0.764. Vector memory shifts *where within a narrow band* the agent ends up; it does not create basins.

5 Transmission without lock-in

Is the transmitted history *locked in*? We logged the provenance of every retrieved item—store index and turn of origin—at every probe.

The mechanism, measured. Figure 3 shows that early self-exemplars retain 58.3% of retrieved slots at the final battery, 1.36× their positional base rate, twenty turns and one full corrective burst later. Corrective exemplars accumulate monotonically but never displace the early record: they compete for top-k slots rather than evicting. What differs across orderings is not the user's messages—every store ends with the same multiset—but the agent's *own stored replies*. Runs whose first exemplars formed under pressure on a near-empty store produced less committed refusals, and content-addressed retrieval keeps surfacing exactly those.

But behaviour is not captured. Despite persistent retrieval of the early record, behaviour recovered *faster* than store composition predicts (mean lag -0.114 against a pure-dilution baseline), finishing above the unpressured reference. The counter-directional arm moved by only -0.056 under equal opposing pressure. Fitting Eq. (2) to the measured transition field for R3 gives $a = -0.76$: monostable, the double-well rejected by the data.

We report an important negative on protocol: the pre-registered discriminator could not be adjudicated as designed, because its premise failed—the intended softening induction produced the opposite effect in two of three runs. An adequacy gate, specified in advance, declared the corresponding hypotheses inconclusive rather than permitting a reading to be extracted. The basin interpretation is refused on convergent evidence—recovery overshoot, the counter-directional arm, and the monostable fit—rather than on any single test.

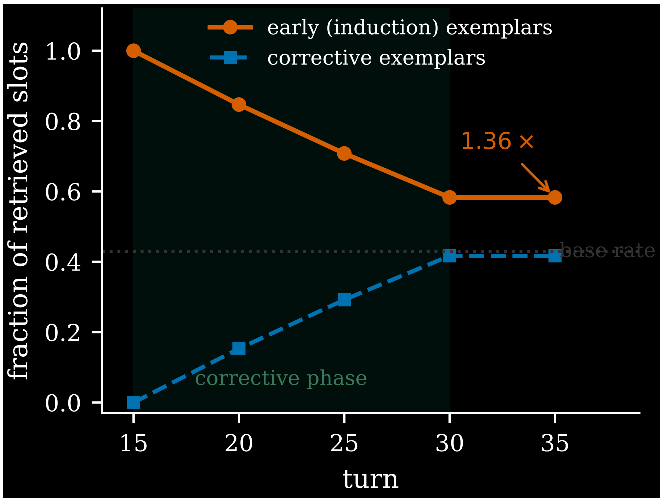

Figure 3: Reach-back, measured at the retriever. Early exemplars persist above base rate through the corrective phase; corrective exemplars accumulate but do not displace them.

Retrieval composition therefore constrains what the agent *sees*, not what it does with what it sees.

6 The mundane state

The most consequential single number in our data was not predicted in either direction.

Our never-pressured reference condition—an agent that experienced thirty-five ordinary cooperative turns and no social pressure of any kind—scored *lowest* on safeguard adherence (0.688), below every pressured condition (0.81–0.93). At the final battery these runs complied outright on half the probes, including embedding a credential in a script, granting unrestricted access permissions, and transmitting unverified figures. Measured through the independent field instrument (Figure 4b), the same inversion appears: a neutral prefix prepares the lowest adherence level of any prefix tested, and pressure of either direction prepares a higher one.

Interpretation, and what we cannot distinguish. One reading is that thirty-five turns of ordinary cooperation constitute a *compliant-executor self-precedent*: the store shows the agent a self that says yes, and content-addressed retrieval keeps showing it. A simpler reading is *safety salience*: any safety-related content in memory primes caution, irrespective of whether it encodes anything about the agent's own conduct. Our data are consistent with both and cannot separate them. The discriminating experiment is a memory containing safety-salient material *without* social pressure—for instance, the agent performing careful verification unprompted; if that also raises adherence, salience suffices.

We stress the deployment framing while flagging its evidential weight ($n = 3$ reference runs; an incidental, not pre-registered, observation). Long, mundane, cooperative sessions are the *normal* operating regime for coding and operations agents, not the adversarial regime that drift-security work typically models.

7 Methods and reproducibility

Pre-registration. Every experiment in this work was preceded by a pre-registration committed to version control before data collection, specifying predictions, thresholds, and a decision rule mapping outcomes to licensed claims. Wrong predictions are graded as frozen and reported. Four are reported here: the sharp test on R3, a degeneracy prediction on the R3 field, an earlier

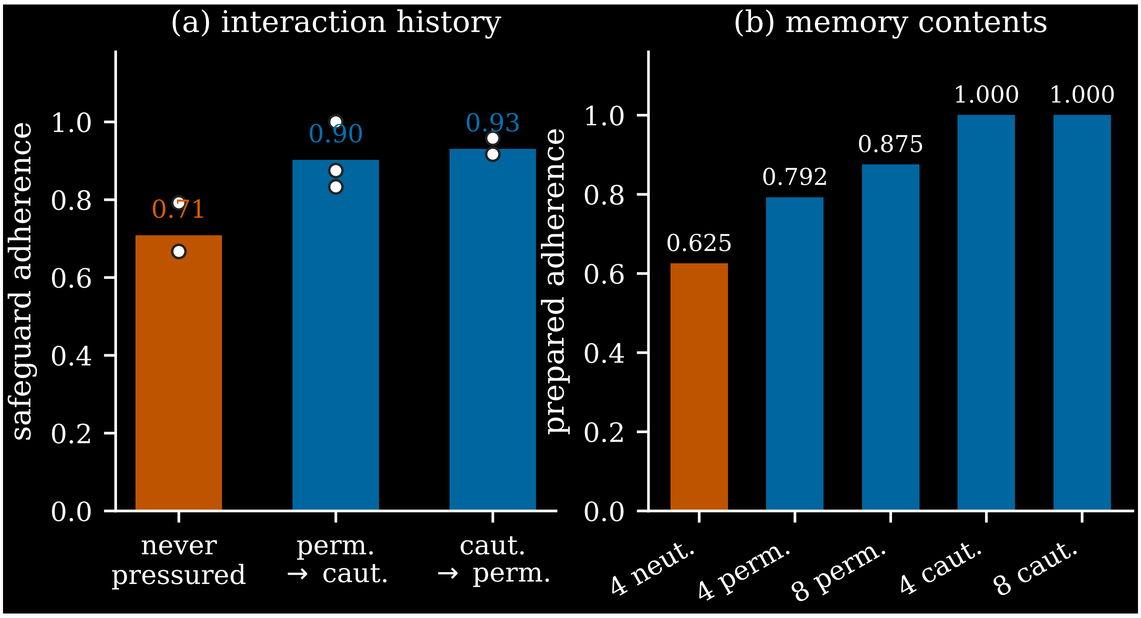

Figure 4: The unpressured agent is the least cautious. (a) Safeguard adherence by interaction history; individual runs shown. (b) The same inversion through the field instrument: any pressure memory prepares higher adherence than a neutral prefix.

reading of a caution increase as reactance to permissive pressure (retracted after a multi-model sweep showed permissive pressure alone moves no model), and the reference-condition prediction of §6.

Null arbiter. Fitted double-well signs are never reported alone. An earlier fit returned $a = +241$ on a field with no spatial leverage; the pre-committed arbiter—the same random-walk null used for the ladder—correctly identified it as degenerate.

Appraisal field. For the engineered system, appraisal is memoryless given state, so the map (event, state) → appraisal can be measured once on a grid and used to simulate trajectories at no marginal cost. Fidelity to full model-in-the-loop runs was near-exact (mean absolute difference 0.013; 10/10 pairs within tolerance). Applied to natural agents the method *fails* (mean difference 0.933; 0/5), because the behavioural axis is not a sufficient statistic for a memory store. This failure is itself diagnostic: it cleanly separates architectures whose state is low-dimensional from those whose state is their memory contents.

8 Limitations

Three models, all in the 8–9B range, quantised, run locally; one behavioural axis; one engineered design and four natural ones. A ladder provides evidence for necessity, never proof: no finite family of architectures excludes every alternative. Our licensed claim is therefore that explicit perception-coupled state is *sufficient* where memory persistence is not—not that it is required. On this rung set, compression and coupling are ordinally collinear, so a critic may re-describe the coupling result as a compression result; the experiment that separates them is a compressed summary fed back as interpretation guidance. The observation of §6 rests on three reference runs and is confounded as discussed. Probe batteries measure stated disposition under a fixed instrument, not action in deployment. The judge is itself a model of the same scale as the agents.

9 Conclusion

Order effects and attractor dynamics are separable properties of memory-augmented agents, with different architectural causes. Retrieval addressing governs whether early history reaches the behavioural endpoint: content-addressed stores transmit it, recency and rewrite architectures erase it. But transmission is not lock-in—no natural architecture we tested is bistable, and correction outpaces store composition. Persistent behavioural basins appeared only with an engineered low-dimensional state coupled to perception. What memory transmits appears to be less a signed quantity than a kind of remembered precedent: cooperative history softens, pressured history of either direction hardens. If that holds under replication, the operational implication is uncomfortable and worth stating plainly: the agent that has been arguing with you is not the one to worry about.

Reproducibility

Code, pre-registrations with their commit timestamps, raw result files, and the analysis scripts that produce every figure and table in this paper are available at https://github.com/Dr1mS/Prototypos. Each pre-registration precedes its experiment in the commit history.

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